Pressure Gradient Theory: Deriving Maxwell's Equations and Periodic Table Structure from a Single Geometric Parameter on the FCC Lattice

PGT v7.0 (v5)

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Abstract

Starting from non-backtracking (NB) paths on a face-centered cubic (FCC) lattice, a 12×12 transfer matrix is constructed using a single geometric parameter $\xi = 1 + (1 + \sqrt{2})/60$. The O_h symmetry decomposition yields five irreducible representations mapped to the four fundamental forces and orbital structure [**P**]; in the long-wave limit, the structure uniquely converges to $\Box A = -J$. The observer axiom states that physics is not "unifying four forces" but "confirming that four forces are four projective experiences of one pressure field." Ionization energy ratios match NIST data (50+ points $< 0.5\%$, best 0.005%). The core derivation (Maxwell's equations and IE ratio structure) has zero free parameters; particle mass formula step counts N are postdiction matches [**O**] (§10.7, §14.3). All results are reproducible in under 100 lines of Python in under 1 second.

Keywords: non-backtracking matrix, FCC lattice, O_h point group, Maxwell equation derivation, ionization energy ratios, zero free parameters

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1 Chapter Zero: Author's Conclusions

1. **Even if PGT fails, the master formula retains practical value.** A zero-parameter formula covering most cross-disciplinary calculations at field-dependent precision (chemistry $< 0.5\%$, physical constants $< 0.1\%$, some domains incomplete). In an era of rising energy demands and escalating computational costs, a single formula replacing multi-software workflows has standalone engineering value—especially for underfunded small research groups.

2. **Those physics masters were neither fools nor cranks.** Einstein, de Broglie, Bohm, and 't Hooft insisted on determinism; they were marginalized by the mainstream. The author chose to stand with them.

3. **LLMs are genuinely useful, with prerequisites.** The prerequisites are: correct usage methodology, rigorous verification procedures, and absolutely never letting them lie. Hallucinations still occur frequently in all frontier models (Claude Opus 4.6, Gemini 3.0 Deep Thinking, GPT-5 Thinking). LLMs are tools, not oracles.

4. **Why publish.** The author originally had no intention of publishing. The realization came later: since determinism implies a unique most-probable solution, hiding it serves no purpose.

2 Chapter One: Problem Statement

2.1 1.1 Motivation

Quantum mechanics achieves extraordinary computational precision, yet its standard interpretation holds that the underlying level is probabilistic, without mechanical causation. Einstein, de Broglie, Bohm, 't Hooft, and others at various times all believed a deterministic substructure might exist. The author chose to stand on this side.

Using Gemini Deep Thinking to decompose the physical constants table, a vacuum pressure of $\sim 10^{47}$ Pa was extracted. Initially suspected to be an error, verification confirmed that QFT/QED already considers the vacuum non-empty (zero-point energy, Casimir effect), and Einstein held the same view.

Prior to this, the author spent one day analyzing experimental mechanisms across disciplines (accelerators, spectrometers, electrochemical cells, torsion balances), discovering that all experimental data ultimately measures force $= -\nabla P$. A pressure field requires a medium; the densest-packing fundamental unit is the regular tetrahedron, which carries the silver ratio $\delta_S = 1 + \sqrt{2}$ and cannot independently fill space (angular gap $\approx 7.35°$, Boerdijk–Coxeter helix), uniquely leading to FCC.

The next week was spent on mathematical unification: using dimensionless constants (α^{-1}, $\sin^2\theta_W$, G) and NIST ionization energies as hard constraints, with cross-disciplinary puzzles (why α^{-1} is prime, why the proton is stable) and discipline-specific challenges as anchoring points to constrain predictions. During this process, some hits turned out to be coincidences and others had rigorous geometric origins—distinguishing the two became the core methodology of subsequent work.

Then one month of rigorous geometric argumentation: building the transfer matrix on FCC, O_h decomposition yielding five irreps, confirming one by one that four forces are projections of the same matrix. BCC ($|d_m|^2$ not uniform), HCP (light speed direction-dependent [**P**]), SC ($|\lambda_{\max}|$ too small [**V**]), and random packing (no long-range order) were all excluded. 33 specific hypotheses were negated and recorded. The parameter is uniquely determined by the Pell equation: $\xi_0 = 1 + \delta_S/60$, not a free parameter.

2.2 1.2 The Questions of This Paper

This paper presents the core results of this reverse engineering, focusing on three independently verifiable questions:

(i) Can Maxwell's equations be derived from NB paths on an FCC lattice—without assuming gauge invariance?

(ii) Can the same structure simultaneously explain the precise IE ratios of the periodic table—with zero additional parameters in the core derivation?

(iii) What is the observer's position in this framework—does the unification problem need to be redefined?

For (i), our answer is affirmative: 9-section complete derivation, symmetry theorems **[P]** lock the equation structure, uniquely converging to Maxwell in the long-wave limit. For (ii), 50+ ratio hits $< 0.5\%$, with perturbation tests confirming structural significance. For (iii), the observer itself is a T_{1u} closed vortex—four forces need not be "unified"; they are already four projections of the same matrix.

2.3 1.3 Grading of Claims

This paper strictly distinguishes the following grades (see Chapter 12 Methodology):

- **[P]** (Proven): Algebraic/group-theoretic theorem valid $\forall\xi$, complete derivation chain
- **[V]** (Verified): Numerically confirmed, reproducible
- **[O]** (Observed): Formula hit is precise but derivation has gaps
- **[C]** (Candidate): Framework exists but lacks quantitative detail
- [×] (Negated): Explicitly excluded, permanently retracted

3 Chapter Two: Literature Review and PGT's Position

3.1 2.1 Non-Backtracking Matrices and the Ihara Zeta Function

PGT's mathematical core—the transfer matrix of non-backtracking (NB) paths—is not invented from thin air. Hashimoto (1989) introduced the NB adjacency matrix in graph theory to study zeta functions of finite graphs [1]. Ihara (1966) earlier defined the related zeta function for discrete subgroups of p-adic linear groups [2]. Bass (1992) gave the determinant formula for the Ihara zeta function, connecting NB matrix spectra to graph structure [3].

PGT's relationship to and differences from this tradition:

- *Common ground:* PGT's transfer matrix $T(\xi)$ is essentially the NB adjacency matrix of FCC's local structure, plus phase weights. $\mathrm{Tr}(T^N)$ counts the phase-weighted number of N-step NB closed paths—precisely the core object of the Ihara zeta function.
- *Difference:* Standard NB matrix elements are 0 or 1 (topology). PGT introduces the phase $\exp(i\theta_{jk}\xi)$, making matrix elements complex. This single modification turns a combinatorial object into a physical one.

3.2 2.2 Lattice Field Theory

Wilson (1974) pioneered lattice gauge theory [4], quantizing gauge fields on discrete lattices to study quark confinement. Wilson's work demonstrated that discrete structures can produce continuous physics—but his lattice is a hypercubic lattice in Euclidean space, which breaks Lorentz invariance at finite lattice spacing and requires taking the continuum limit.

Dialogue between PGT and lattice gauge theory:

- *Same spirit:* Discrete structure can produce continuous physics.
- *Different route:* Wilson starts from the SM gauge group $SU(3) \times SU(2) \times U(1)$ and places the continuous theory on a lattice. PGT reverses this—starting from the FCC lattice's O_h symmetry group and deriving the force structure.
- *Confinement mechanism:* Wilson requires strong-coupling expansion and numerical simulation to argue for confinement. In PGT, T_{2g}'s flat band (zero dispersion **[P]**) directly gives "does not propagate = confinement"—a one-step derivation, not a numerical result.

3.3 2.3 Lorentz Invariance on Discrete Structures

"Can a discrete lattice produce Lorentz invariance?" is an old question in lattice field theory. PGT provides a concrete mechanism: three-layer protection (see Chapter 8). This has parallels with the causal sets research direction [5], but the starting points differ—PGT does not start from general relativity but from the lattice's algebraic structure.

3.4 2.4 PGT's Distinctive Feature

Compared to all above work, PGT's core difference is: **zero free parameters.** Lattice gauge theory has coupling constants; the Standard Model has 25+ parameters; string theory has $\sim 10^{500}$ vacua. PGT has only $\xi_0 = 1 + (1 + \sqrt{2})/60$, and ξ_0 is not free—it is uniquely determined by the Pell equation $\delta_S^2 = 2\delta_S + 1$.

4 Chapter Three: Axiom System

PGT is built on four axioms.

Axiom 1 (Discrete Paths and Topological Projection, A0)
There exists an FCC close-packed lattice with lattice constant ℓ_0. Each site has 12 nearest neighbors. The backtracking path $A \to B \to A$ has zero net displacement—topologically equivalent to the null operation—and is therefore projected out. After projection, each step has $12 - 1 = 11$ effective directions; nonzero matrix elements $= 12 \times 11 = 132$.

Axiom 2 (Phase Cost Per Step, A1)
The turning angle θ_{jk} (from incoming direction j to outgoing direction k) produces a phase:

$$T_{jk} = \begin{cases} \exp(i \cdot \theta_{jk} \cdot \xi) & j \neq -k \\ 0 & j = -k \text{ (NB projection)} \end{cases}$$

Axiom 3 (Geometric Uniqueness, A2)
δ_S is uniquely determined by the face-to-face stacking geometry of regular tetrahedra:

$$\delta_S^2 = 2\delta_S + 1 \implies \delta_S = 1 + \sqrt{2}$$

Geometric origin: the regular tetrahedral dihedral angle θ satisfies $\cos\theta = 1/3$, so $\cot(\theta/2) = \sqrt{2}$, giving $\delta_S = 1 + \cot(\theta/2)$—the Pell equation's characteristic root equals one plus the half-dihedral cotangent. The denominator 60 corresponds to the tetrahedral face angle 60°: in Axiom A1, $\alpha = e^{i\pi\xi/3}$, where $\pi/3$ radians $= 60°$ is the face-to-face angle of FCC tetrahedral interstices [**P**]. Thus $\xi_0 = 1 + \delta_S/60$ means "phase cost = base angle (60°) + Pell perturbation (δ_S°)," $\xi_0 = 1.040236\ldots$

Regarding octahedra: Pure regular tetrahedra cannot fill 3D space (angular gap $\approx 7.35°$, Boerdijk–Coxeter helix). FCC is the unique close-packed solution obtained by adding regular octahedra to fill the angular gap (§6.3). The existence of octahedra does not modify δ_S—the causal arrow is reversed: *because* $\delta_S \neq 0$ (geometric frustration of tetrahedra), octahedra are *needed* to close space. δ_S defines the frustration; octahedra are its consequence.

Axiom 4 (Coherent Superposition, A3)
Observables = phase-weighted sum of all NB paths after topological projection:

$$Z(N) = \mathrm{Tr}(T^N) = \sum_{\text{all } N\text{-step NB closed paths}} \exp(i \times \text{accumulated phase})$$

Input inventory: Geometric constants: 0 (all of δ_S, 12, $\sqrt{2}$ determined by axioms). Physical anchors: 3 ($c, \hbar, \bar{m}_e$ = unit-system constants). Free parameters: **0**.

5 Chapter Four: Observer Axiom

5.1 4.1 Core Statement

Observer = axiom. The observer's irrep Γ_{obs} defines its entire observable range through $\Gamma_{\text{obs}} \otimes \Gamma'$.

Physics = T^N projected onto Γ_{obs}. Different $\Gamma_{\text{obs}} \to$ different physics.

5.2 4.2 Human Physics = T_{1u} Projection

Humans are composed of electrons (T_{1u} closed vortices) and measure with photons (T_{1u} propagating waves). Therefore human physics = T_{1u} self-reference.

Object Γ'	Experience	Mainstream theory
T_{1u}	Direct observation (absorb/emit)	QED
A_{1g}	Deflected (trajectory bending)	GR
T_{2g}	Confined (energy trapped)	QCD
E_g	Deformed (tidal)	GR tensor
T_{2u}	Nearly imperceptible	Weak force

Gravitational channel structure: PGT's gravity is not a scalar theory. A_{1g} (dim = 1) provides the static Poisson potential ($\nabla^2\Phi = -4\pi\rho$); E_g (dim = 2) provides dynamic gravitational waves (two polarizations $h_+, h_\times$, exactly matching GR's traceless symmetric tensor structure). $E_g \otimes E_g \ni E_g$ **[P]** guarantees gravitational self-coupling (nonlinearity = Einstein's essential feature); $A_{1g} \otimes \Gamma = \Gamma$ **[P]** guarantees the equivalence principle. The explicit derivation of the full Einstein field equations is not yet complete **[C]**, but all components are in place. Light deflection = T_{1u} photon trajectory deflected in A_{1g} gradient ($A_{1g} \otimes T_{1u} = T_{1u}$ **[P]** guarantees coupling); GR's 2× factor requires the E_g tensor contribution—this quantitative derivation awaits completion **[C]**.

5.3 4.3 Reversal of the Unification Direction

Mainstream has spent 50 years on: four theories → find unification.
PGT's direction: one matrix (already unified) → confirm four theories are its T_{1u} projections.

Not "unifying four forces"—but "confirming four forces are four observational experiences of the same pressure field."

5.4 4.4 Corollaries

- Why do IE ratios reach 0.005%? Because the measurement tool (T_{1u}) and measured structure come from the same T matrix; ratios cancel systematic corrections.
- The NIST Atomic Spectra Database = T_{1u}'s selfie.
- The periodic table = electrons decoding their own channel structure.

6 Chapter Five: Analytic Eigenvalues and Identities

6.1 5.1 O_h Decomposition

The FCC point group O_h (48 elements) decomposes the 12-dimensional bond representation into five irreducible representations:

$$\Gamma_{12} = A_{1g}(1) \oplus T_{1u}(3) \oplus T_{2g}(3) \oplus E_g(2) \oplus T_{2u}(3)$$

Projection operators $P_\Gamma = (d_\Gamma/48)\sum_g \chi_\Gamma(g)^* D(g)$ verified: $P_\Gamma^2 = P_\Gamma$, $\sum P_\Gamma = I$, $P_\Gamma P_{\Gamma'} = 0$ (all at machine precision).

6.2 5.2 Three Fundamental Phases and Five Analytic Formulas

Theorem 6.1 (Analytic eigenvalues, $\forall\xi$ **[P]**). *Let $\alpha = e^{i\pi\xi/3}$, $\beta = e^{i\pi\xi/2}$, $\gamma = \alpha^2 = e^{i2\pi\xi/3}$. Then:*

$$\lambda_{A_{1g}} = 1 + 4\alpha + 2\beta + 4\gamma \tag{1}$$
$$\lambda_{T_{1u}} = 1 + 2\alpha - 2\gamma \tag{2}$$
$$\lambda_{T_{2g}} = 1 - 2\beta \tag{3}$$
$$\lambda_{E_g} = 1 - 2\alpha + 2\beta - 2\gamma \tag{4}$$
$$\lambda_{T_{2u}} = 1 - 2\alpha + 2\gamma \tag{5}$$

Derivation sketch: Each irrep Γ has standard basis functions $f_\Gamma(\hat{d})$. The eigenvalue satisfies $(Tf_\Gamma)_j = \lambda_\Gamma \cdot f_\Gamma(\hat{d}_j)$. For T_{1u}: take $f_x(\hat{d}) = d_x$, sum over neighbors at $j = (+1, +1, 0)/\sqrt{2}$. Key result: the two 90° neighbors have opposite x-components, canceling exactly—β vanishes. Thus T_{1u} contains no β: electromagnetism does not participate in orthogonal jumps **[P]**. Similarly, T_{2g} ($f_{xy} = d_x d_y$ type) has nonzero basis functions only for 90° neighbors—color force sees only 90° jumps **[P]**.

6.3 5.3 Numerical Results

irrep	dim	$\lvert\lambda\rvert$	arg(°)	N_{close}	Physics
A_{1g}	1	8.836	+87.14	4.1	Gravity
T_{1u}	3	3.071	+2.44	147	Electromagnetism
T_{2g}	3	2.292	−60.56	5.9	Color force
E_g	2	1.789	−52.46	6.9	Tidal force
T_{2u}	3	1.076	−173.0	2.1	Weak force

Note: $N_{\text{close}} = 360°/|\arg(\lambda)|$ is the bare closure step count. T_{1u}'s $N_{\text{bare}} = 147$ (full 3D helical steps) differs from the experimentally observable $N_{\text{phys}} = \alpha^{-1} = 137$ (prime): the latter is the effective step count seen by the observer from a specific projection angle k_0, with k_0 uniquely determined by the non-resonance condition (N being prime) **[O]**.

6.4 5.4 Core Identities

17 proven identities (P1–P16 + D2), all rigorously valid $\forall\xi$ **[P]**. The most essential:

- P1: $\sum_\Gamma \dim(\Gamma)|\lambda_\Gamma|^2 = 132$ (normal matrix $+|T_{jm}| = 1$ **[P]**)
- P4: $\lambda_{T_{1u}} + \lambda_{T_{2u}} = 2$ (weights are exact negatives **[P]**)
- P5: $|\lambda_{T_{2g}} - 1| = 2\ \forall\xi$ (T_{2g} lies on the circle centered at 1, radius 2 **[P]**)
- P9: $\beta^2 = \alpha\gamma = e^{i\pi\xi}$ (three angle steps share a common source **[P]**)

7 Chapter Six: Geometric Theorems

7.1 6.1 FCC Uniqueness [P]

Theorem 7.1. *Among close-packed lattices, only FCC simultaneously satisfies Axioms A0–A3.*

Five independent exclusions of HCP: (1) NB projection inconsistency (HCP antiparallel cos $= -0.833 \neq -1$); (2) no BC helix (HCP dihedral $\to$ Axiom A2 has no source); (3) slow-mode sign reversal; (4) T_{1u} triple degeneracy broken (D_{6h}: $T_{1u} \to A_{2u}(1) + E_{1u}(2) \to$ direction-dependent speed of light $\to$ Lorentz violation); (5) threshold robustness collapse. SC (simple cubic) excluded at ξ_0: $|\lambda_{\max}(\text{SC})| = 4.06 \ll 8.84$ **[V]** (ξ-scan finds SC$\geq$FCC at $\xi \approx 2.53$, so SC exclusion holds only near ξ_0, not yet an $\forall\xi$ algebraic proof).

7.2 6.2 Tetrahedral Tube Theorem [P]

Theorem 7.2. *Spherical harmonics Y_ℓ^m arise from FCC tetrahedral interstice geometry, not from Coulomb potential symmetry.*

Three-step proof: (1) FCC tetrahedral interstices have an exact inscribed sphere (T_d-invariant) **[P]**; (2) standing waves on this sphere = Y_ℓ^m decomposed under T_d **[P]**; (3) $|\lambda|$ ordering = orbital ordering: $|A_{1g}| > |T_{1u}| > |T_{2g}| > |E_g| > |T_{2u}|$ corresponds to $s > p > d > f$ **[P]**.

Causal arrow reversal: mainstream says "Coulomb potential → symmetry → Y_ℓ^m"; PGT says "FCC interstice geometry → tube sphere → Y_ℓ^m → Coulomb potential is downstream."

7.3 6.3 Dual-Tube Structure Theorem [P]

The minimal complete unit of one FCC step $\ell_0 = 4$ regular tetrahedra + 1 regular octahedron, volume ratio exactly 1 : 1.

Lattice vs. interstice roles: The T matrix (Axioms A0–A1) is constructed on bonds between lattice sites—pressure propagates along lattice-site connections, with O_h symmetry (including inversion). But physical entities (vortices/electrons) = standing-wave modes of the pressure field in interstice tubes (§6.2): tet interstices have T_d boundary (no inversion), oct interstices have O_h (with inversion).

$O_h \to T_d$: T_{1u} (EM) and T_{2g} (color) both map to T_2 → photon and gluon are indistinguishable under T_d. The coexistence of two interstice types makes inversion physically meaningful—it is precisely the oct interstice's inversion symmetry that separates propagating modes (T_{1u}, ungerade) from bound modes (T_{2g}, gerade) **[P]**.

7.4 6.4 Grand Shell Theorem [P]

Theorem 7.3. *The 14 interstice sites (8 tet + 6 oct) surrounding one FCC lattice point decompose under O_h as:*

$$\Gamma_{14} = 2A_{1g} \oplus 2T_{1u} \oplus T_{2g} \oplus E_g \oplus A_{2u}$$

Capacity: $14 \times 2 = 28$ (= Ni's atomic number, exact count **[P]**).

Key structure: T_{2g} appears only in tet ($n = 1, 0$), E_g only in oct ($n = 0, 1$), T_{2u} does not appear—consistent with the analytic formulas' tube separation.

NN coupling domain cutoff: Extending to the fifth shell (including body-centered octahedral S2b (8 sites) = O_h mirror of S1 **[P]**), totaling 94 interstice sites ($8 + 6 + 24 + 8 + 24 + 24$), all with NN vertex count ≥ 1. Beyond S6, NN= 0, decoupled from the central nucleus's T matrix **[P]**. Capacity $94 \times 2 = 188 > 118$ (known elements). S6 (32 sites) is the first layer beyond the cutoff; $94 + 32 = 126$ (= nuclear magic number, see §9.6) **[O]**.

7.5 6.5 Shell Effective Field Closed-Form Theorem [P]

Theorem 7.4. *Define the shell effective field $F(S_n) = \sum_\Gamma w_\Gamma^{(S_n)} \cdot \lambda_\Gamma$. Then:*

$$F(S_1) = 1 + 2\alpha, \quad F(S_2) = 1 + 2\alpha + \beta, \quad F(S_3) = 1 + \alpha$$

*valid $\forall \xi$ **[P]**. The coefficient of α equals that shell's 60° contact face count.*

In $F(S_3)$, the β and γ terms exactly cancel—S3 sees neither octahedral bridges nor cross-group connections; its pressure environment is purely tetrahedral **[P]**.

S3's refractive index $\arg(F)/\arg(\alpha) = 1/2$ (half-angle formula $\arg(1 + e^{i\theta}) = \theta/2$, exact **[P]**). The S2↔S3 impedance mismatch (reflectivity $R = 0.39$) is the largest among three shell pairs **[V]**—corresponding to the large IE jump at $n = 2 \to n = 3$. Physical interpretation: **the FCC lattice is the medium; shielding = impedance mismatch; IE = energy cost of removing one standing-wave mode.**

8 Chapter Seven: From Lattice to Maxwell—Complete Derivation

This is the paper's core chapter. Below we show how to derive the complete Maxwell equations from Axioms A0–A3. The derivation has 9 sections: symmetry theorems are all [**P**]; the transition from discrete differences to continuous PDEs relies on the long-wave approximation (§7.4, §7.6).

8.1 7.1 One-Step Propagation Equation [P]

$$\psi_\Gamma(N+1) = \lambda_\Gamma \left[\psi_\Gamma + \frac{a^2}{12}\nabla^2\psi_\Gamma\right] \tag{6}$$

where $a^2/12 = \ell_0^2/6$ is identical for all five irreps [**P**].

8.2 7.2 Global Phase Factorization [P]

$$M_\Gamma(k) = \lambda_\Gamma(0) \times R_\Gamma(k) \tag{7}$$

$R_\Gamma(k)$ is a real symmetric matrix (FCC inversion pairs all sites as $\pm d \to$ real matrix [**P**]). Therefore all eigenvalues of M_Γ have constant argument $\forall k \in \mathrm{BZ}$ [**P**].

8.3 7.3 Three Special Properties of T_{1u}

1. Constant arg $\to$ dispersion-free $\to$ massless [**P**]
2. Trace isotropy $\to$ group velocity direction-independent: $\mathrm{Tr}(P_{T_{1u}} \cdot T(k))/3$ is an O_h invariant (group theory [**P**]); physical corollary: "group velocity is isotropic" [**V**]
3. $R(k)$ longitudinal/transverse ratio $= 2:1 \to$ two polarizations [**P**]

8.4 7.4 From Discrete to Wave Equation [P structure + long-wave approximation]

The fundamental equation is the one-step propagation of §7.1 (direct consequence of T^N); the wave equation is downstream. Steady state ($\psi(N+1) = \psi(N)$) = dynamical fixed point (time average), not "no time."

In the one-step equation, $|\lambda_{T_{1u}}| = 3.071 > 1$: the $k = 0$ uniform mode grows in amplitude each step—this is pressure field background, not physical EM waves. Physical waves = deviations from the background.

Why A_{1g}'s exponential growth does not swamp T_{1u}: $|A_{1g}|/|T_{1u}| \approx 2.88$; at large N, A_{1g} exponentially dominates—but this does not destroy T_{1u} structure, for three reasons. (1) O_h orthogonality [**P**]: T matrix is strictly block-diagonal in irrep basis, as is T^N; energy transfer between channels is exactly zero—A_{1g} growth stays in A_{1g}, does not flow into T_{1u}. (2) Uniform growth elimination: all five channels have $|\lambda| > 1$ background growth; the physical field is defined as the deviation; after elimination, each channel evolves independently. (3) Steady-state isolation: matter = Helmholtz steady-state solution ($\psi(N+1) = \psi(N)$); the steady-state condition exactly eliminates the growth factor—the existence of steady-state solutions does not depend on the magnitude of $|\lambda|$.

Derivation in three steps:

1. *Eliminate uniform growth:* Renormalize $A \to A/|\lambda_{T_{1u}}|^N$, extract k-dependent deviation. Transverse effective propagator: $|\lambda_{T_{1u}}|(1 - k^2/8)$ (§7.2 factorization + §6.2 $R(k)$ transverse eigenvalue $4 - k^2/2$).
2. *Define physical field = consecutive two-step difference:* $\delta\tilde{A}(N) \equiv \tilde{A}(N+1) - \tilde{A}(N)$, eliminating $k = 0$ residual. Take one more difference to get second-order $\delta^2\tilde{A}/\tau^2$, where $\tau = \ell_0/c$.
3. *Factorization locks propagation speed uniqueness:* Spatial term $\propto k^2$ (i.e., ∇^2); temporal term $\propto 1/\tau^2$. Factorization theorem ensures the ratio is independent of $|k|$ [**P**]; trace theorem ensures independence of k direction [**P**]. Combined $\to T_{1u}$ propagation speed is the unique isotropic, dispersion-free constant [**P**].

$$\boxed{\frac{\partial^2 A_\perp}{\partial t^2} = c^2\,\nabla^2 A_\perp} \tag{8}$$

Status of c: Axioms A0–A3 define spatial structure and stepping rules but do not independently define time. $t_0 \equiv \ell_0/c$ maps discrete steps to continuous time—$c = \ell_0/t_0$ is definitional consistency (tautology), not a derivation. The genuine **[P]** content is "there exists a unique isotropic dispersion-free propagation speed"; its numerical value is fixed by unit-system anchors $\hbar, m_e$.

Precision note: The above is the discrete one-step equation in the long-wave approximation ($k\ell_0 \ll 1$). The wave equation's *structure* (dispersion-free + isotropic + linear + transverse) is **[P]**, but the exact equality truncates $O(k^4\ell_0^4)$ terms (= dimension-6 Lorentz violation, $\sim 10^{-44}$ at optical wavelengths).

8.5 7.5 Bound/Radiation Decomposition [P]

$$\psi_{T_{1u}}^{\text{total}} = \psi_{\text{bound}} + \psi_{\text{rad}} \tag{9}$$

Steady state ($\psi(N{+}1) = \psi(N)$) = matter's T_{1u} component. Non-steady = radiation. Substituting the decomposition into the one-step equation:

$$\psi_{\text{rad}}(N{+}1) = \lambda_{T_{1u}}\left[\psi_{\text{rad}} + \frac{a^2}{12}\nabla^2\psi_{\text{rad}}\right] - \delta\psi_{\text{bound}}$$

where $\delta\psi_{\text{bound}} = \psi_{\text{bound}}^{C(N+1)} - \psi_{\text{bound}}^{C(N)}$ = matter configuration change.

8.6 7.6 Source Maxwell Equation

$$\boxed{\Box A^\mu = -J^\mu} \tag{10}$$

J^μ is not an assumption—it is the algebraic consequence of $\delta\psi_{\text{bound}}$ (matter's T_{1u} configuration change) under the long-wave approximation. The continuous form of the source equation depends on the long-wave approximation; the mechanism itself (bound $\to$ radiation) is an exact corollary of the discrete one-step equation **[P]**.

At rest, no radiation ($\delta\psi = 0$); uniform motion, no radiation (steady state invariant under Lorentz boost); acceleration radiates ($\delta\psi \neq 0$)—not three rules, but three special cases of the same equation **[P]**.

8.7 7.7 Charge Conservation [P]

$\partial_\mu J^\mu = 0$ follows from three independent routes: (1) P1 pressure conservation ($\sum \dim|\lambda|^2 = 132$ **[P]** $+O_h$ orthogonality forbids inter-irrep energy transfer $\to T_{1u}$ channel share independently conserved)—this is the primary proof route; (2) NB topology (closed vortex winding number is integer, pair production required $\to$ net vortex number conserved); (3) linear structure ($T_{1u} \otimes T_{1u} \not\ni T_{1u}$ $\to$ no self-coupling; a necessary but not sufficient condition alone—requires channel isolation from route (1)).

8.8 7.8 Lorenz Condition [P]

$\partial_\mu A^\mu = 0$ is not a gauge choice—it is the geometric fact of $R(k)$'s longitudinal/transverse splitting. Far field is purely transverse.

8.9 7.9 Derivation Chain Summary

$$\xi \to T(\xi) \to O_h \to T_{1u} \text{ block} \to M = \lambda R \to \text{disp.-free} + \text{isotropic} + \text{transverse} \to \Box A = -J$$

Symmetry theorems (factorization, trace, linearity, L/T ratio) lock the wave equation's structure [**P**]. The transition from discrete differences to continuous PDEs requires $k\ell_0 \ll 1$, truncating $O(k^4\ell_0^4)$. Rigorous statement: the discrete one-step equation uniquely converges to Maxwell's equations in the long-wave limit—"uniquely converges" is [**P**] (locked by symmetry, no other possible form); the exact equality holds for $k\ell_0 \ll 1$.

9 Chapter Eight: Lorentz Invariance

9.1 8.1 Three-Layer Protection Mechanism [P]

Theorem 9.1 (Three-layer Lorentz protection)**.**

1. *A_{1g} isotropy: singlet isotropy lemma **[P]**—exact isotropy at k^2 order; anisotropy begins at k^4 (bare value $b/|C_2| = 0.023$).*
2. *T_{1u} trace theorem (observer-equivalent isotropy): $\mathrm{Tr}(P_{T_{1u}} \cdot T(k))/3$ is an O_h invariant, depending only on $|k|$, not direction **[P]**. Dimension-4 (k^2 anisotropy) = exactly zero; dimension-5 (k^3) = exactly zero (O_h includes inversion → CPT). Observer's Lorentz violation starts at dimension-6 (k^4), further suppressed $57\times$ **[V]**.*
3. *Vortex finite-size suppression: closed vortex (N steps) Lorentz violation $\delta_{\mathrm{LV}} \leq 0.023/N^2$; all known particles have $N \geq 73 \to \delta_{\mathrm{LV}} < 5 \times 10^{-6}$ **[P]**.*

Observer's Lorentz violation (not bare): $\delta_{\mathrm{LV}}^{\mathrm{obs}} \approx 0.137 \times k^4$, where $k = E/E_0$. Visible light (2 eV): $\sim 10^{-44}$; electron (0.5 MeV): $\sim 10^{-22}$; proton (1 GeV): $\sim 10^{-9}$. Hughes–Drever-type experiments constrain dim-4/5 coefficients at 10^{-20}–10^{-30}—PGT is exactly zero at these orders [**P**]. The dim-6 electron-scale prediction $\sim 10^{-22}$ is at the boundary of current experimental sensitivity—a directly testable prediction.

UHECR constraint: $E_0 = 85.3$ GeV is not a UV cutoff. In PGT, particles = closed vortices with internal phase (k_{internal}). External acceleration to 14 TeV changes the center-of-mass velocity, not the vortex's internal Bloch k—just as accelerating a He atom does not change its internal electron states. UHECR protons (A_{1g}, $N = 74$) sit at $k_{\mathrm{internal}} \approx 0$; GZK threshold margin $\sim 10^4$. LHC energy $\gg E_0$ poses no contradiction.

9.2 8.2 Comparison with Lattice Field Theory

Wilson's (1974) lattice gauge theory breaks Lorentz invariance at finite lattice spacing and relies on the continuum limit ($a \to 0$) for recovery [4]. PGT's FCC lattice ensures isotropy through group theory (O_h's high symmetry) rather than taking a continuum limit. The difference: the hypercubic lattice has O_h only as a subgroup, while FCC's O_h is the largest symmetry group among close-packed lattices.

10 Chapter Nine: Experimental Verification

10.1 9.1 IE Ratio Method: Principle

Algebraic layer: If IE $\propto |\lambda_\Gamma|^{f(N)} \times$ (common factor containing $\hbar, c, N, g$), then the ratio of two IEs cancels the common factor → pure $|\lambda|$ ratio [**P**].

Physical layer: Multi-electron atoms' IE includes electron–electron interactions (repulsion, exchange-correlation)—highly nonlinear and without simple closed forms. That ratios still achieve 0.005% precision means many-body corrections approximately cancel proportionally between numerator and denominator. The candidate explanation is T_{1u} self-reference (measurement tool and

measured structure from the same T matrix), but a rigorous mathematical proof of many-body cancellation does not yet exist [**O**].

Therefore: ratios canceling unit-system constants is algebraic fact [**P**]; ratios canceling many-body corrections is observational phenomenon [**O**]. Precision is 10–100× higher than absolute formulas.

10.2 9.2 Selected Hits (Deviation $< 0.01\%$) [V]

Ratio	$\lvert\lambda\rvert$ ratio	Precision	Source
Ne IE_2/IE_6	T_{2g}/A_{1g}	0.007%	NIST ASD
IE(F)/IE(H)	T_{2g}/E_g	0.007%	NIST ASD
Ni/As IE_1 ratio	E_g/T_{2g}	0.005%	NIST ASD
D(Cl–Cl)/D(O–O)	E_g/T_{2u}	0.008%	CRC Handbook

Cross-disciplinary scanning also found bond-energy ratio hits (e.g., D(H–H)/IE(F)), but bond energies have 0.1–1% source variation due to reference-standard differences (D_0 vs D_e vs D_{298}); precision claims require caution. The first three rows come from NIST data [6], independently verifiable; total 50+ ratio hits $< 0.5\%$.

10.3 9.3 Five Control Ratios Decode the Periodic Table [V]

The periodic table IE_1 sawtooth is alternately controlled by five $|\lambda|$ ratios: E_g/T_{1u} (s-pairing jump, 0.7%), T_{2g}/E_g ($s^2 \to p^1$ jump, 0.3%), E_g/T_{2g} (p-shell decrease, 0.15%), T_{1u}/T_{2g} (cross-period same-group, 0.26%), $\Pi(E_g)/\Pi(T_{2g})$ ($d^5\,s$ pairing, 0.37%).

10.4 9.4 Perturbation Test

Shifting each $|\lambda|$ by $\pm 1\%$ and re-running the ratio scan: hits collapse from 29 to 9 (3.9σ). The hits are structural, not random coincidences.

10.5 9.5 Physical Constants (Closed-Form, Each with Derivation Gaps)

Quantity	PGT formula	Deviation	Grade
$\sin^2\theta_W$	$2\xi/9$	-0.024%	[**O**]
G	Closed-form (see appendix)	$+0.009\%$	[**O**]
m_μ/m_e	$\delta_S^6 \exp(\arg\lambda_{T_{1u}})$	-0.072%	[**O**]

Each [**O**] has a specific gap: the denominator 9 of $\sin^2\theta_W$ is not yet derived from first principles; G's δ_S^{84} lacks a group-theoretic derivation.

10.6 9.6 Cross-Disciplinary Extensions [O]/[V]

Nuclear binding energy ratios: B(Ne-20)/B(Ca-40) $= |T_{2u}|/|T_{2g}|$ (deviation 0.0008%, AME2020 data [**O**])—among the most precise hits in cross-disciplinary scanning. Chemical IE ratios are dominated by T_{2g}/E_g (electrons sensing color-force shielding); nuclear binding energies are dominated by T_{2u}/T_{2g} (weak/color coupling in nuclear forces).

Nuclear magic numbers: 8, 28, 126—three exact hits to FCC shell structure numbers (8 = S1 site count [**P**]; 28 = S1+S2 capacity [**P**]; 126 = 94 + 32 = NN coupling domain + S6 [**O**]). The numbers 2 and 20 have no natural correspondence (PGT weakness [×]).

Aufbau geometric ordering: The bare ordering of $V_{\text{eff}} = |\lambda_\Gamma|^2 \times w_\Gamma/r^2$ exactly reproduces $1s < 2s < 2p < 3s < 3p$ filling order [**V**]. The $4s/3d$ inversion comes from T_{2g} self-shielding correction [**V**]. The $S_4 \equiv S_5$ projection degeneracy (V_{eff} difference 5%) is the geometric root of $n = 4$ level crossing [**V**].

11 Chapter Ten: What PGT Currently Cannot Do

Honesty is the core of the methodology.

1. **Absolute energies of arbitrary atoms/molecules:** PGT gives ratio structure; it does not replace DFT/*ab initio*.

2. **Precise quark masses and hadron spectrum:** T_{2g}'s flat band gives the confinement mechanism (pressure waves do not propagate [**P**]), but the precise mapping between $\dim(T_{2g}) = 3$ and the Standard Model $SU(3)$'s 8 gluon degrees of freedom has not been established. PGT does not presuppose it must reproduce the $SU(3)$ gauge group—O_h's irrep structure is a derived result, not an input—but a complete hadron spectrum calculation requires a many-body theory for the T_{2g} channel, currently lacking [**C**].

3. **CKM/PMNS matrices:** Quantitative formulas for three-generation mixing not established.

4. **Dark matter candidates:** Framework exists (non-T_{1u} vortices), but no quantitative prediction.

5. **Complete field equation assembly:** Maxwell complete [**P**]; Einstein materials ready but assembly not done [**C**].

6. **Derivation gaps in all [O] items:** $\sin^2\theta_W$ denominator 9, G's δ_S^{84}, m_μ formula's causal chain.

7. **Particle mass formula:** $E(\Gamma, N) = |\lambda_\Gamma^N - 1| \times E_0$ matches some mass ratios at [**O**] precision ($m_\pi/m_e = |T_{1u}|^5$, deviation $+0.04\%$; $m_\mu/m_e = \delta_S^6 \exp(\arg \lambda_{T_{1u}})$, deviation -0.07%; $m_n - m_p$ deviation -0.001%). But each N is a postdiction (matching experimental values after the fact), not a prediction. Exponents (e.g., the 5 in $|T_{1u}|^5$) lack first-principles derivation. $m_p/m_\mu = |A_{1g}|$ deviation -0.50%—the worst among all hits. Perturbation test ($\pm 1\%$): hits collapse from 29 to 9 (3.9σ) [**O**].

8. **Spin-1/2 and Fermi statistics:** T_{1u} is O_h's 3D vector representation (integer spin), but electrons are spin-1/2 fermions. PGT has a candidate mechanism—FCC dual tetrahedra ($T^+ \oplus T^-$, 4 faces each) require 4π traversal = two 2π rotations to return [**C**]—but rigorous derivation is missing. Pauli exclusion has a group-theory version (coexistence matrix strictly diagonal [**P**]: $\Gamma_1 \otimes \Gamma_2 \ni A_{1g} \Leftrightarrow \Gamma_1 = \Gamma_2$), but explicit construction of anti-commutation relations is incomplete. This does not affect the paper's three core results (Maxwell derivation and IE ratios do not depend on spin structure), but affects completeness of particle classification and mass spectrum.

9. **Algebraic derivation of IE ratios $\approx |\lambda|$ ratios:** The steady-state Helmholtz field equation ($\nabla^2\psi + \mu^2\psi = S$, $\mu^2 = 12(\lambda - 1)/(a^2\lambda)$ [**P**]) provides the channel through which $|\lambda|$ enters IE (via Green's function and shielding), but the complete algebraic derivation from field equation to "IE ratios exactly equal $|\lambda|$ ratios" does not yet exist [**O**]. Perturbation test (3.9σ) and MC joint matching ($> 4\sigma$) exclude random coincidence, but the causal chain has a gap.

10. **Radiation cancellation for uniform motion on a discrete lattice:** §7.6 proves that steady state ($\delta\psi = 0$) does not radiate [**P**]. Uniform motion is invariant under continuous Lorentz boost, but the discrete FCC lattice lacks exact continuous translational symmetry—a vortex advancing step-by-step has local $\delta\psi_{\text{bound}} \neq 0$, requiring a global cancellation mechanism (lattice Bremsstrahlung suppression) to make macroscopic net radiation zero. Algebraic proof of this mechanism: [**C**].

11. **Ontological isolation of eigenvalue growth:** $|\lambda_\Gamma| > 1$ (all five channels) means the one-step equation $\psi(N+1) = T\psi(N)$ is non-unitary. §7.4 implicitly eliminates uniform growth through wavefunction differencing (standard lattice field theory operation); O_h orthogonality guarantees no inter-channel mixing [**P**]. But the ontological question—"what does the pressure field background's exponential growth physically represent?"—is currently answered as "steady-state solutions exist regardless of $|\lambda|$ magnitude" (§7.4); the connection from pressure field dynamics to thermodynamic equilibrium has not been established [**C**].

11.1 10.1 Lessons from Negated Items

During the development of this theory, 33 items were explicitly negated [×] and 6 downgraded. Each negation is a methodological victory—shrinking the search space.

- **D1 arg identity [×]:** A specific linear combination of five $\arg(\lambda_\Gamma)$ matched at ξ_0 to 0.0009°, but ξ-scanning revealed complete collapse at other ξ. Lesson: holding only at ξ_0 does not imply universality—ξ-scanning is a necessary condition for [**P**].
- 12ε **hypothesis [×]:** Hypothesized that the 0.3% deviation cluster shares a common correction factor 12ε. Numerical scan ruled out: per-step corrections differ by 10× across formulas. Lesson: "patterns" from small data sets may be artifacts.
- $c_3 = 1/6$ **[×]:** A specific guess for a shielding coefficient. ξ-scan negated.

12 Chapter Eleven: Falsification Conditions

Any one of the following experimental results would directly negate PGT:

1. The same ℓ_0 yields contradictory values when computed from G, σ, and H_0.
2. The photon is measured to have nonzero mass $> 10^{-18}$ eV (in PGT, photon masslessness is an algebraic result [**P**]).
3. The $\alpha(Q)$ running curve is incompatible with T_{1u} dispersion.
4. The equivalence principle is violated beyond the irrep mixing level at high-precision measurement.
5. A fourth generation of fermions is discovered (in PGT, the regular tetrahedron has only 4 faces − 1 axis = 3 generations [**P**]).

13 Chapter Twelve: Methodology

13.1 12.1 Five-Level Proof Grading

Grade	Definition
[**P**]	Algebraic proof $\forall\xi$, complete derivation chain. ξ-scan ($[0.01, 10]$, 2000+ points) zero violations
[**V**]	Numerically confirmed but not rigorously proven
[**O**]	Formula hit but derivation has gap
[**C**]	Conceptual framework, lacks quantitative detail
[×]	Definitively wrong, permanently retracted

13.2 12.2 Core Principles

1. Ratios > formulas: ratios cancel dynamical corrections, 10–100× higher precision
2. ξ-scanning is a necessary condition for [**P**]: holding only at ξ_0 is not a theorem
3. External anchoring prevents closed loops: low internal RMS $\neq$ correct
4. Exhaustive scanning > guessing formulas
5. Perturbation test: $|\lambda|$ shift $\pm 1\%$ causes hit collapse = structural
6. Honest labeling: [**O**] is not [**P**]; downgrade rather than overclaim
7. Negative results are progress: every [×] shrinks the search space

14 Chapter Thirteen: LLM Usage Declaration

14.1 13.1 Models Used

Model	Primary use
Claude Opus 4.6 (Anthropic)	Main workhorse: derivation expansion, numerical verification, document writing, red-team review
Gemini 3.0 Deep Thinking / Thinking / Pro (Google)	Physical constants table decomposition (discovered 10^{47} Pa), cross-verification
GPT-5 Thinking / 5.1 Thinking (OpenAI)	Cross-verification, alternative derivation routes
Grok 3 / 4.2 (xAI)	Cross-verification

14.2 13.2 Division of Labor

- **Human (BlackJakey):** Theoretical insights, directional judgments, formula selection, core concepts (observer axiom, ratio method principle, $-\nabla P$ starting logic, tube theorem causal arrow reversal)
- **AI (all models above):** Computational verification, derivation expansion, document writing, red-team review

14.3 13.3 LLM Contribution Level

- Conceptual layer: ∼0%. All core physical insights from the human author.
- Derivation layer: ∼40%. LLMs expand derivation details and perform numerical verification.
- Writing layer: ∼70%. Document formatting and language polishing primarily by LLMs.
- Review layer: ∼90%. Red-team review (multiple rounds, including L0–L5 six-layer checks) executed by Claude.

14.4 13.4 Adversarial AI Cross-Verification (AACV)

This theory underwent multiple rounds of red-team review:

- v6.9.3.2 RT4: Fatal 0, Required 0, Noted 2, Passed 7
- v6.9.4: Fatal 0, Required 1, Noted 3
- v6.9.7.1: Fatal 0, Required 3, Noted 5
- v6.9.8.2: All mathematics exact (17 items verified)

Red-team specification (v6.9.8.1) includes L0 ontological classification layer, specifically detecting the logical error of "treating an analytical tool as a physical entity"—a lesson learned from actual errors in version v6.9.8.

14.5 13.5 Known Limitations of LLMs

All models (including frontier versions) have erred in: (1) hallucinations—generating nonexistent citations or formulas; (2) basic arithmetic—even Claude Opus 4.6 and GPT-5 Thinking occasionally miscalculate simple multiplication; (3) Claude tends to write "computable and verifiable" results into documents while skipping "conceptual directions"; Gemini has produced intermediate results requiring human re-verification during constants table decomposition. All LLM output was cross-verified by the human author before acceptance.

Formal verification (e.g., Lean, Coq) is a reasonable direction for further rigorization; currently not executed. All **[P]** results in this paper are verifiable by the Python code in Appendix A. Specific algebraic operations (tensor products, irrep decompositions) can be symbolically verified with GAP or Mathematica.

15 Chapter Fourteen: Originality and Novelty Statement

15.1 14.1 Novel Contributions of This Paper

1. **Mapping from NB matrices to physics:** Hashimoto/Ihara/Bass's NB matrix is a purely mathematical object. PGT introduces phase weights and O_h decomposition, mapping it to the structure of physical forces. To the author's knowledge, this mapping is unprecedented.
2. **Complete derivation of Maxwell's equations:** From lattice axioms to $\Box A^\mu = -J^\mu$; symmetry theorems [**P**] lock the equation structure; unique convergence in the long-wave limit.
3. **Observer axiom:** Reframes the unification problem from "unify four forces" to "confirm four forces are projections of the same matrix."
4. **Zero-parameter prediction of IE ratios:** 50+ ratio hits $< 0.5\%$, best 0.005%, with perturbation tests confirming structural significance.
5. **Shell effective field closed-form theorem:** $F(S_n)$ contains only phase factors of contact-face angles [**P**], equating the FCC lattice to a medium—shielding = impedance mismatch; IE = energy cost of removing a standing wave. Nuclear binding energy ratio (B(Ne-20)/B(Ca-40) = $|T_{2u}|/|T_{2g}|$, 0.0008%) and chemical IE ratios arise from the same T matrix structure [**O**].
6. **Complete negation record:** 33 items [×] + 6 downgraded, each with lesson.

15.2 14.2 Distinction from Existing Literature

- vs. lattice field theory: PGT does not assume a gauge group; O_h is a derived result
- vs. string theory: PGT introduces no extra dimensions
- vs. causal sets: PGT's lattice is FCC, not randomly sprinkled points
- vs. numerical relativity: PGT does not discretize known equations on a lattice, but derives equations from the lattice

15.3 14.3 Limitations

The theory's [**O**] results ($\sin^2\theta_W$, G, mass formula) each have derivation gaps. Mass formula step counts N are postdiction (after-the-fact matching), not prediction. The complete Einstein field equation derivation is not finished [**C**]. Rigorous derivation of spin-1/2 and Fermi statistics: [**C**] (§10.8). Algebraic causal chain for IE ratios $\approx |\lambda|$ ratios has a gap [**O**] (§10.9).

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A Appendix A: Verification Code

The following Python code independently reproduces all numerical results in this paper (< 100 lines, < 1 second).

```
import numpy as np
# === Constants ===
dS = 1 + np.sqrt(2)
xi = 1 + dS / 60
alpha = np.exp(1j * np.pi * xi / 3)
beta  = np.exp(1j * np.pi * xi / 2)
gamma = alpha**2
# === Analytic eigenvalues ===
lam = {
    'A1g': 1 + 4*alpha + 2*beta + 4*gamma,
    'T1u': 1 + 2*alpha - 2*gamma,
    'T2g': 1 - 2*beta,
    'Eg':  1 - 2*alpha + 2*beta - 2*gamma,
    'T2u': 1 - 2*alpha + 2*gamma,
}
dims = {'A1g':1,'T1u':3,'T2g':3,'Eg':2,'T2u':3}
for g, l in lam.items():
    print(f"  {g}: |lam|={abs(l):.6f}, "
          f"arg={np.degrees(np.angle(l)):+.4f} deg")
# === Identity checks ===
P1 = sum(dims[g]*abs(lam[g])**2 for g in lam)
P4 = lam['T1u'] + lam['T2u']
P5 = abs(lam['T2g'] - 1)
assert abs(P1 - 132) < 1e-10, "P1 FAIL"
assert abs(P4 - 2)   < 1e-10, "P4 FAIL"
assert abs(P5 - 2)   < 1e-10, "P5 FAIL"
print(f"P1={P1:.10f}, P4={P4:.10f}, P5={P5:.10f}")
# === IE ratio examples ===
IE_F, IE_H = 17.42282, 13.59844
r_exp = IE_F / IE_H
r_pgt = abs(lam['T2g']) / abs(lam['Eg'])
print(f"IE(F)/IE(H): exp={r_exp:.5f}, "
      f"PGT={r_pgt:.5f}, err={((r_pgt/r_exp)-1)*100:+.4f}%")
# === Physical constants ===
sw2 = 2*xi/9
print(f"sin^2 theta_W = {sw2:.6f} "
      f"(PDG 0.23122, err {(sw2/0.23122-1)*100:+.4f}%)")
# === xi scan: P1 = 132 for all xi ===
violations = 0
for xi_t in np.linspace(0.01, 10, 2000):
    a = np.exp(1j*np.pi*xi_t/3)
    b = np.exp(1j*np.pi*xi_t/2)
    g = a**2
    ls = {'A1g':1+4*a+2*b+4*g, 'T1u':1+2*a-2*g,
          'T2g':1-2*b, 'Eg':1-2*a+2*b-2*g,
          'T2u':1-2*a+2*g}
    s = sum(dims[k]*abs(ls[k])**2 for k in ls)
    if abs(s-132) > 1e-10: violations += 1
print(f"xi scan: 2000 pts, violations={violations}")
```

B Appendix B: Symbol Table

Symbol	Definition	Value
δ_S	$1+\sqrt{2}$ (Pell equation)	$2.41421\ldots$
ξ_0	$1+\delta_S/60$	$1.04024\ldots$
α	$e^{i\pi\xi/3}$	60° step phase
β	$e^{i\pi\xi/2}$	90° step phase
γ	$\alpha^2 = e^{i2\pi\xi/3}$	120° step phase
ℓ_0	FCC lattice constant (derived: $\bar{\lambda}_C \times N \times g$)	$2.3136 \times 10^{-18}\,\mathrm{m}$
E_0	$\hbar c/\ell_0$	$85.3\,\mathrm{GeV}$
N_{bare}	Bare closure steps $= 360^\circ/\lvert\arg(\lambda_{T_{1u}})\rvert$	147
N_{phys}	Physical closure steps $= \alpha^{-1}$ (projected, prime)	137

C Appendix C: Supplementary Materials

Complete algebraic derivation manuscripts (including O_h projection operator construction, full-BZ numerical scan data, complete record of 33 negated items, Maxwell derivation 18-step expansion) are archived in the supplementary materials repository. The verification code in Appendix A independently reproduces all numerical results; the supplementary materials provide complete derivation details for peer review.

PGT v7.0 (v5) · Pressure Gradient Theory · Journal Ambitions Contest Submission
Core derivation: zero free parameters · Fully reproducible · Complete source code: Appendix A